

Cybersecurity (F26)

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Course Plan

Lecture	Date	Teacher	Topic
01	2026-02-04	Sani	Introduction & Lab Set-up
02	2026-02-11	Sani	Vulnerability Assessment
03	2026-02-18	Sani	Penetration Testing
04	2026-02-25	Hiraku	Firewalls and IDS
05	2026-03-04	Hiraku	Malware & SQL
06	2026-03-11	Hiraku	Denial of Service
07	2026-03-18	Sani	Social Engineering, Phishing
08	2026-03-25	Sani	Threat Modelling I
09	2026-04-08	Hiraku	Cryptography
10	2026-04-15	Hiraku	Multi-party Computation
11	2026-04-22	Jukka	Privacy, Data Protection, and GDPR
12	2026-04-29	Sani	Threat Modelling II, Revision, & Exam Prep.

Threat Modeling II, Revision, Mock Test

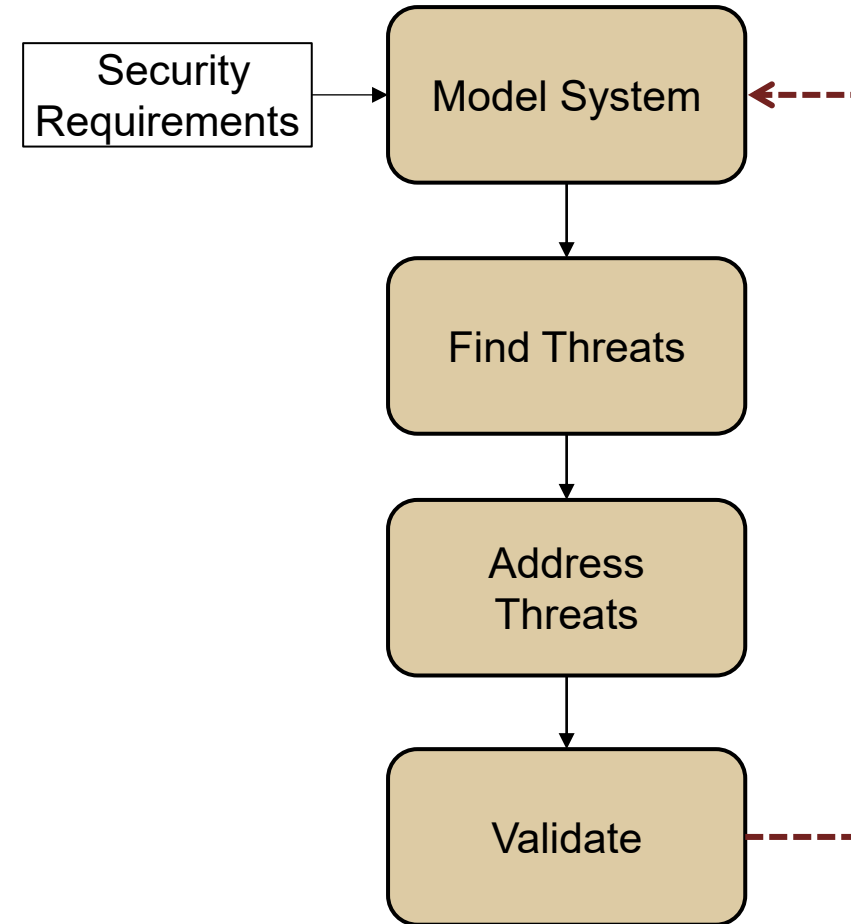
Recall

What is threat modelling?

What are the key questions to ask when you perform threat modeling?

Steps involved in the threat modeling process:

- Creating an application diagram to model the system
- Identify the threats
- Mitigate the threats
- Validate the model to ensure threats are mitigated



Use Case: Kubernetes Threat Modeling

Kubernetes Security

The 4Cs of cloud-native systems

1. Cloud

- Security of the cloud infrastructure

2. Cluster

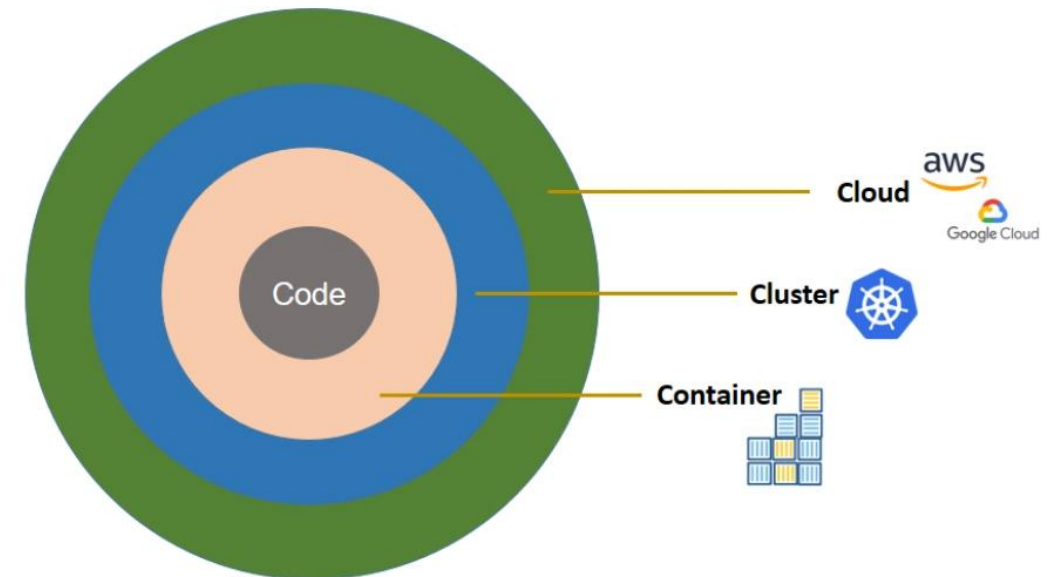
- Kubernetes cluster-level security

3. Container

- Security of the Dockerfile/Containerfile, image, etc

4. Code

- Security of the actual code implementation

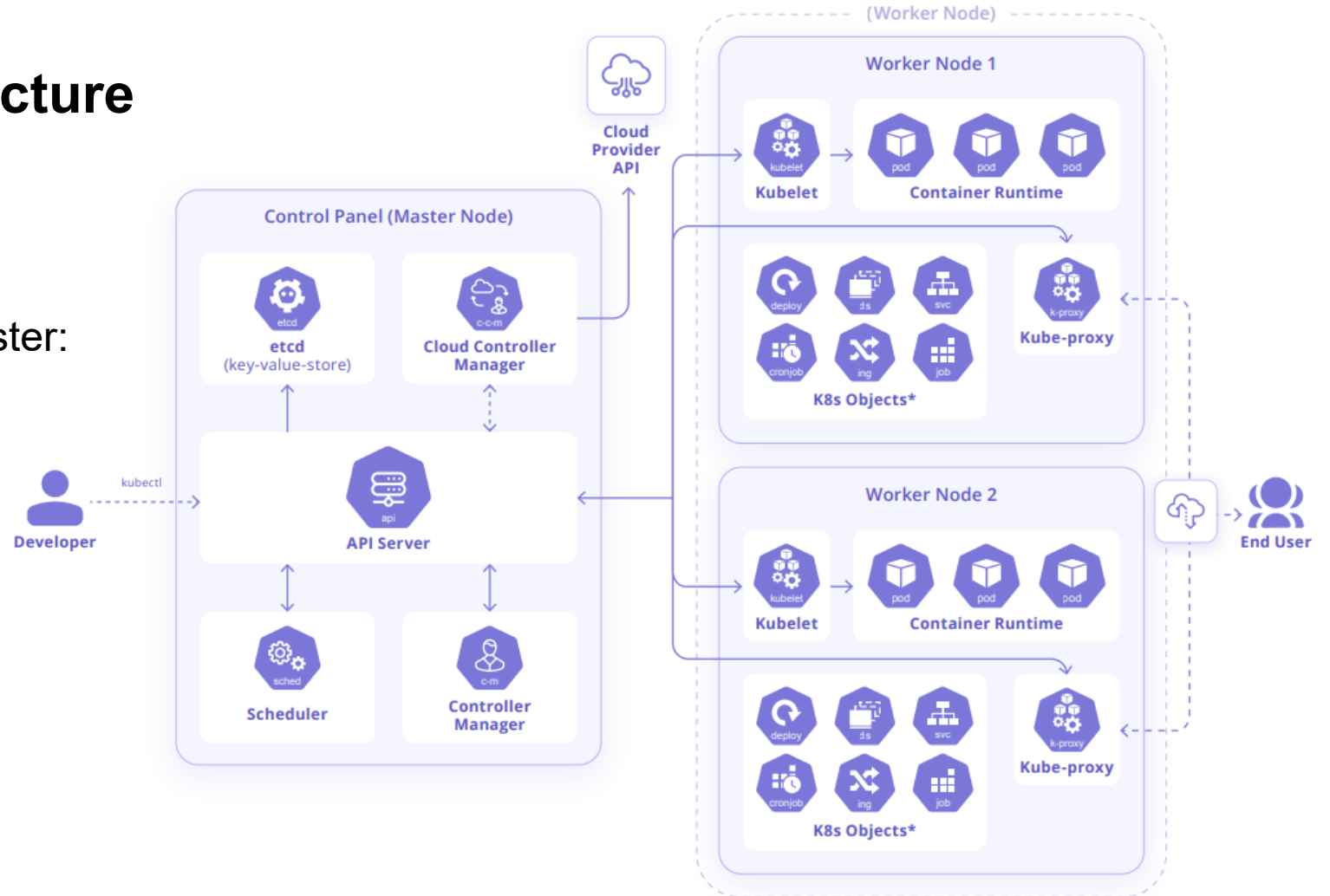


Kubernetes Security

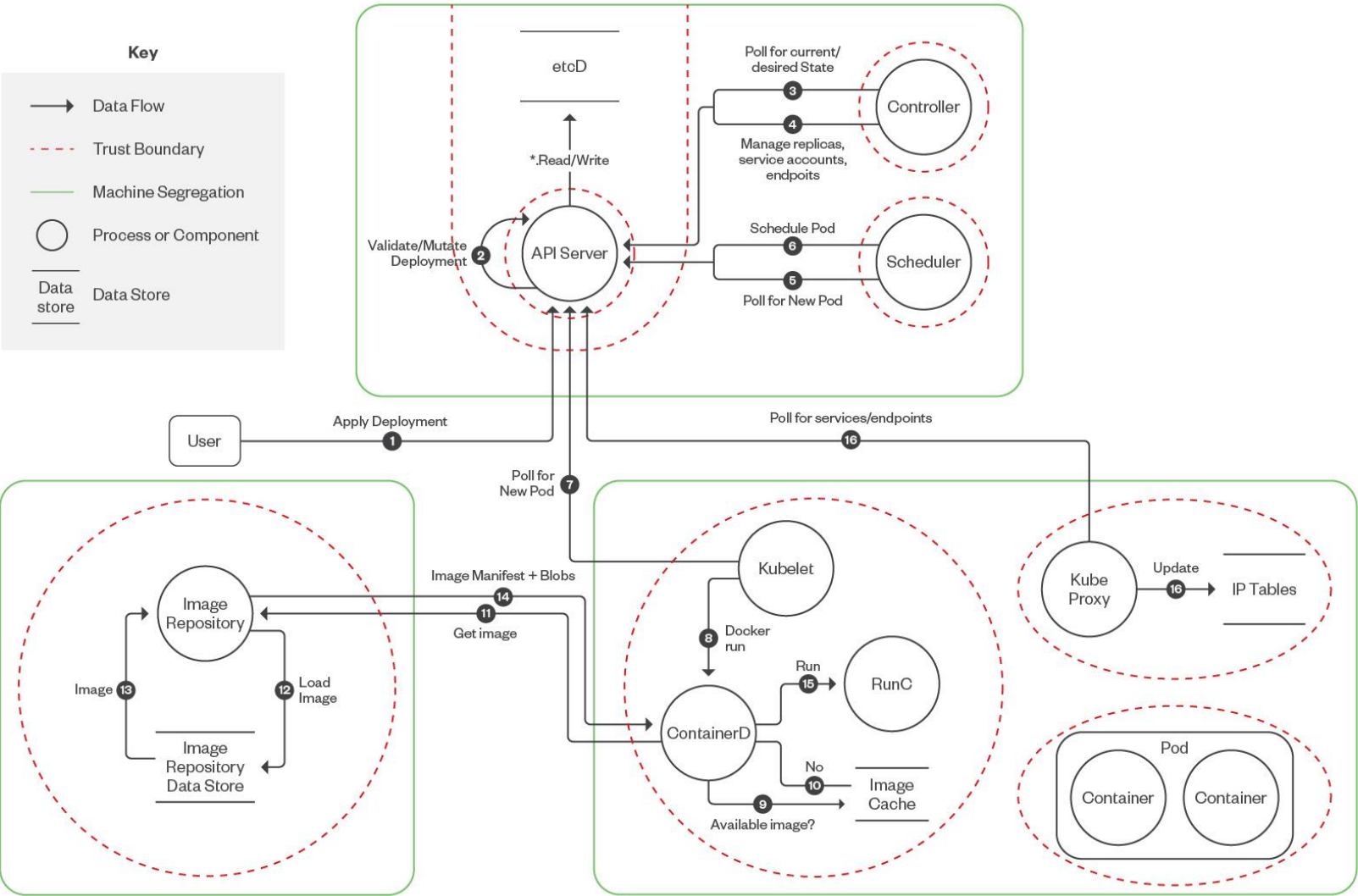
Understanding the Architecture

Main components of Kubernetes cluster:

- Control plane
- Worker nodes



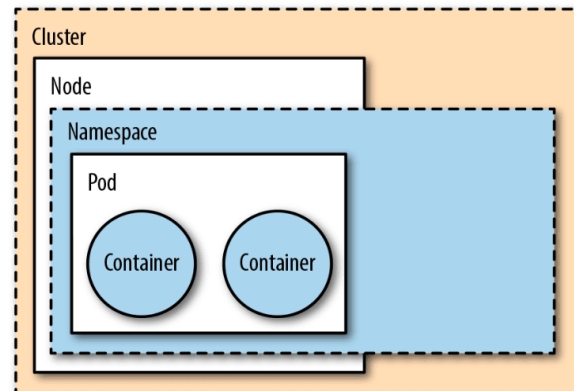
Kubernetes Threat Model



Kubernetes Threat Model

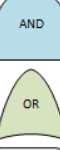
Main Attack Vectors

- **Denial of Service:** Prevent the system from functioning by exhausting resources within the cluster.
- **Compromised Container:** Provides a remote execution point for an attacker
- **Service Token Compromise:** Allows unauthorized access to cluster API.
- **Network Endpoints:** Access to the endpoints if the pods network policy permits.
- **RBAC Issues:** Many attack vectors rely on mis-configuration of RBAC policies.



[Security boundaries](#)

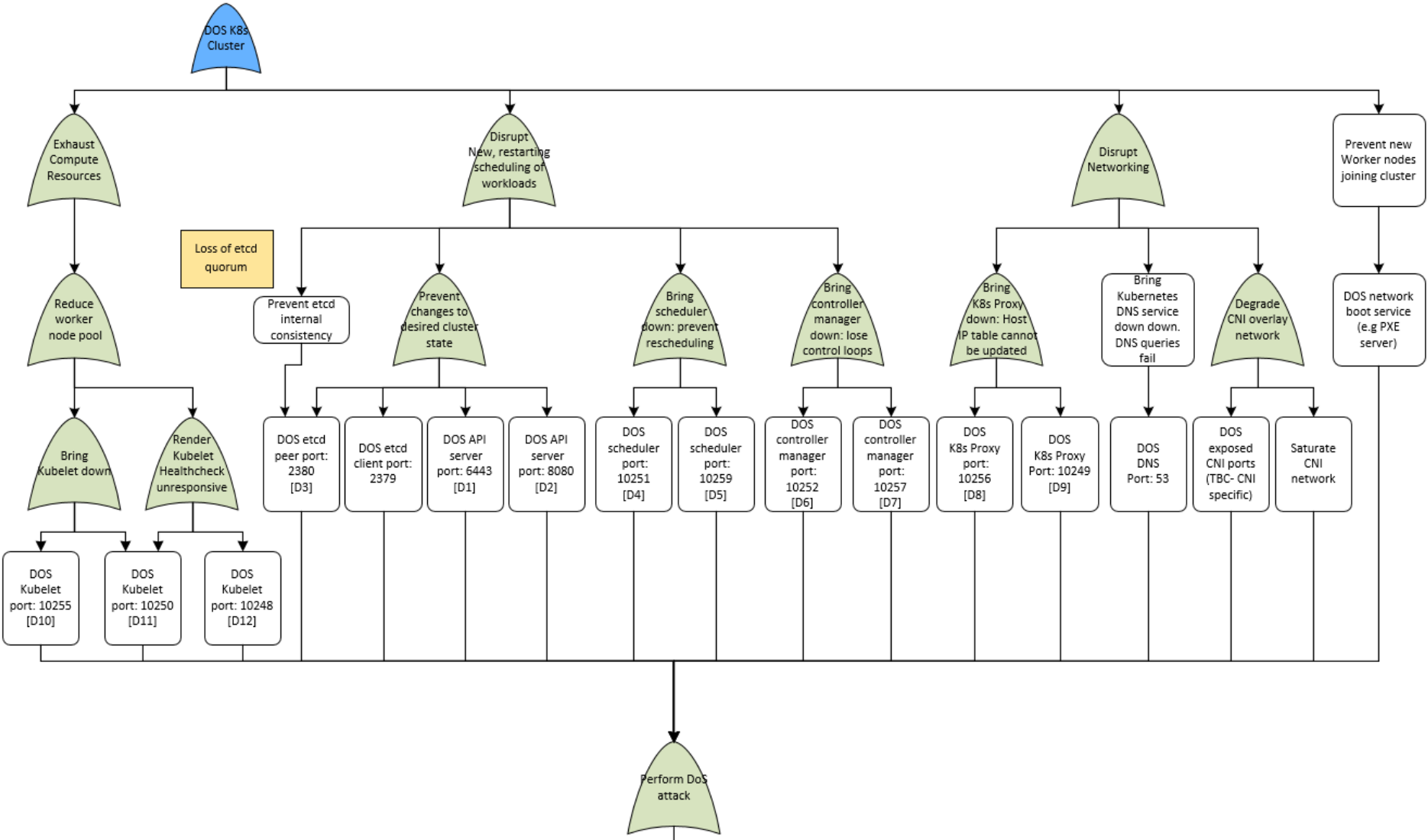
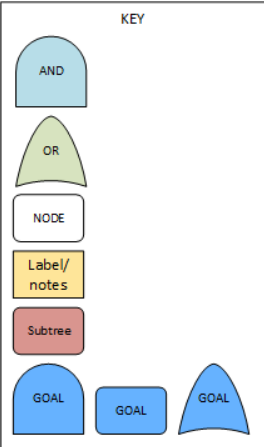
#sdudk

- KEY
- 
- AND
- OR
- NODE
- Label/
notes
- Subtree
- GOAL
- GOAL
- GOAL

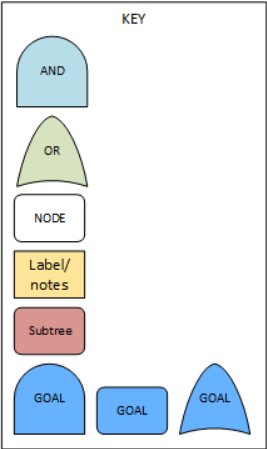


Kubernetes Threat Model

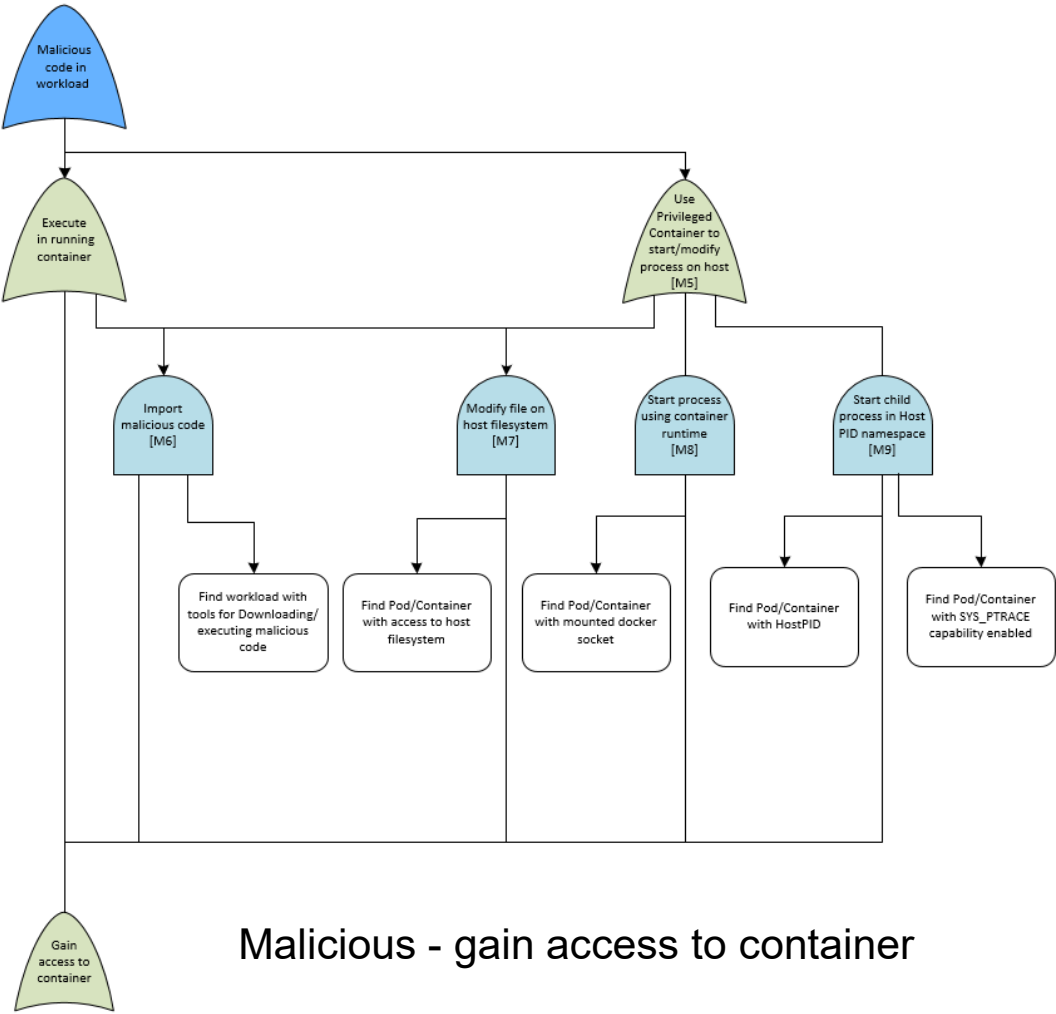
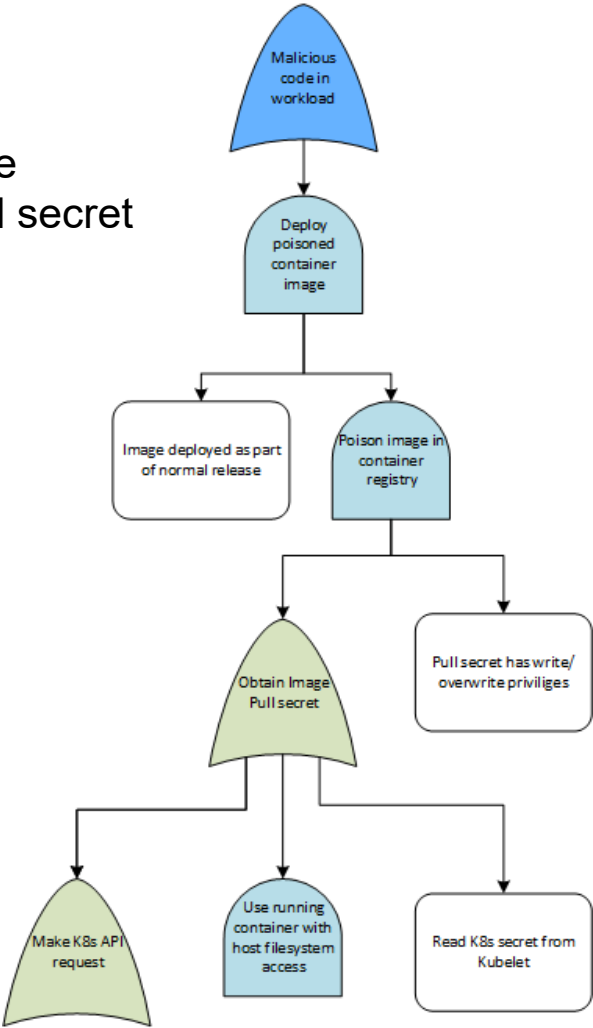
DoS: Right branch



Kubernetes Threat Model



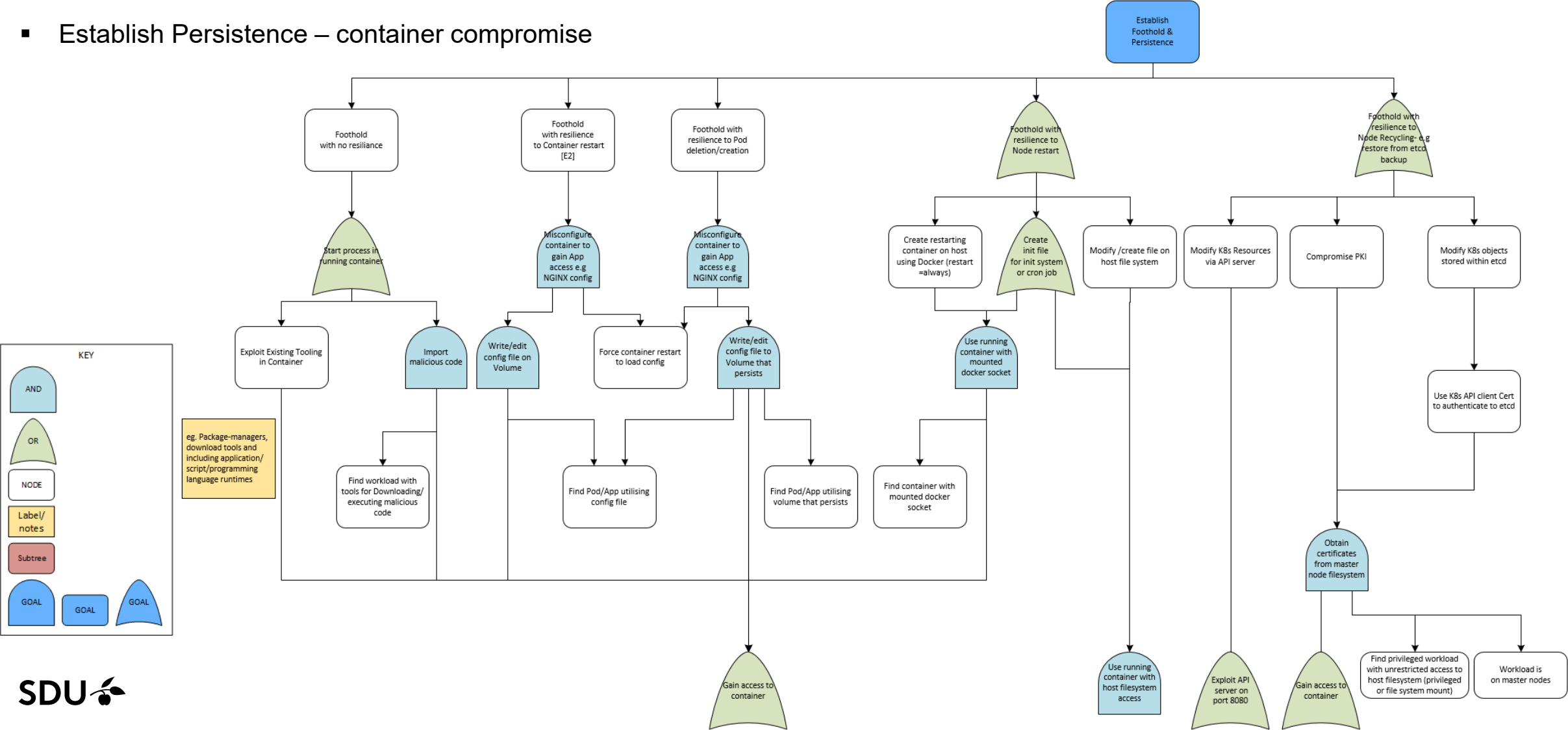
Malicious Code Execution- pull secret



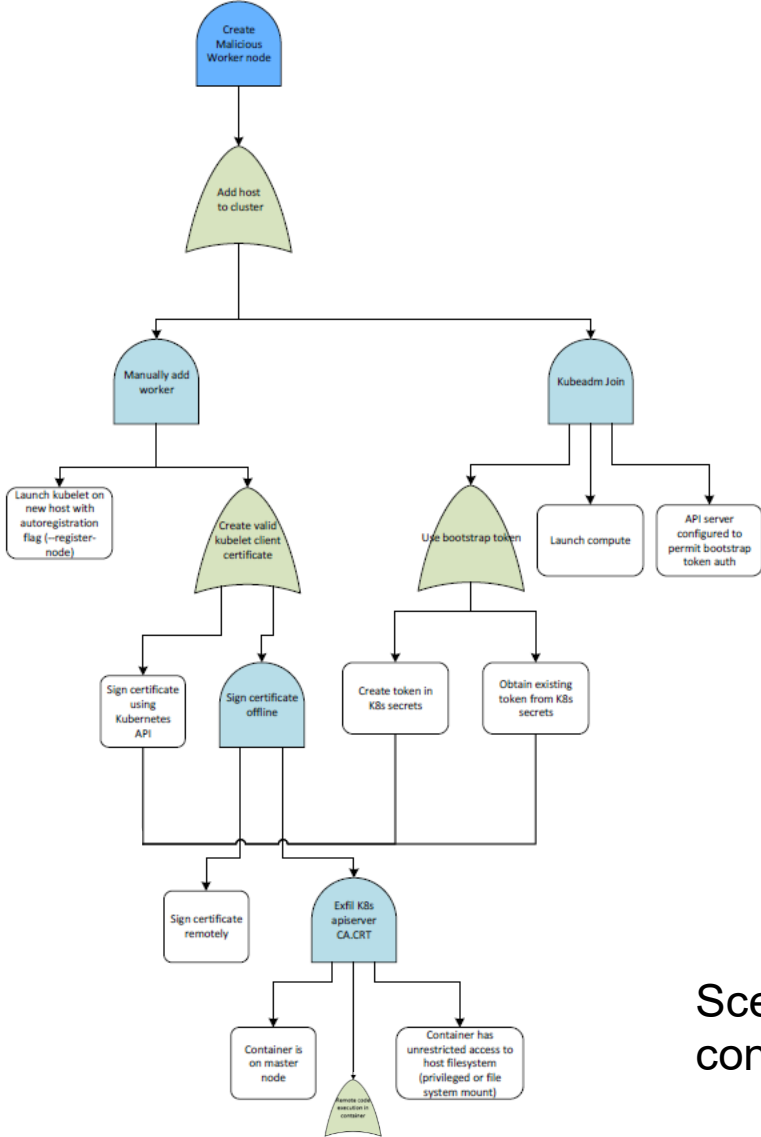
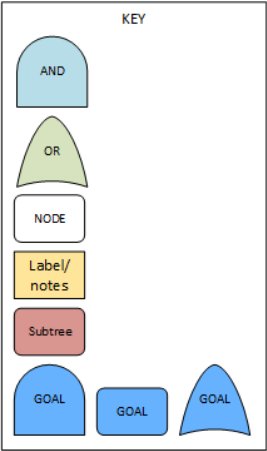
Malicious - gain access to container

Kubernetes Threat Model

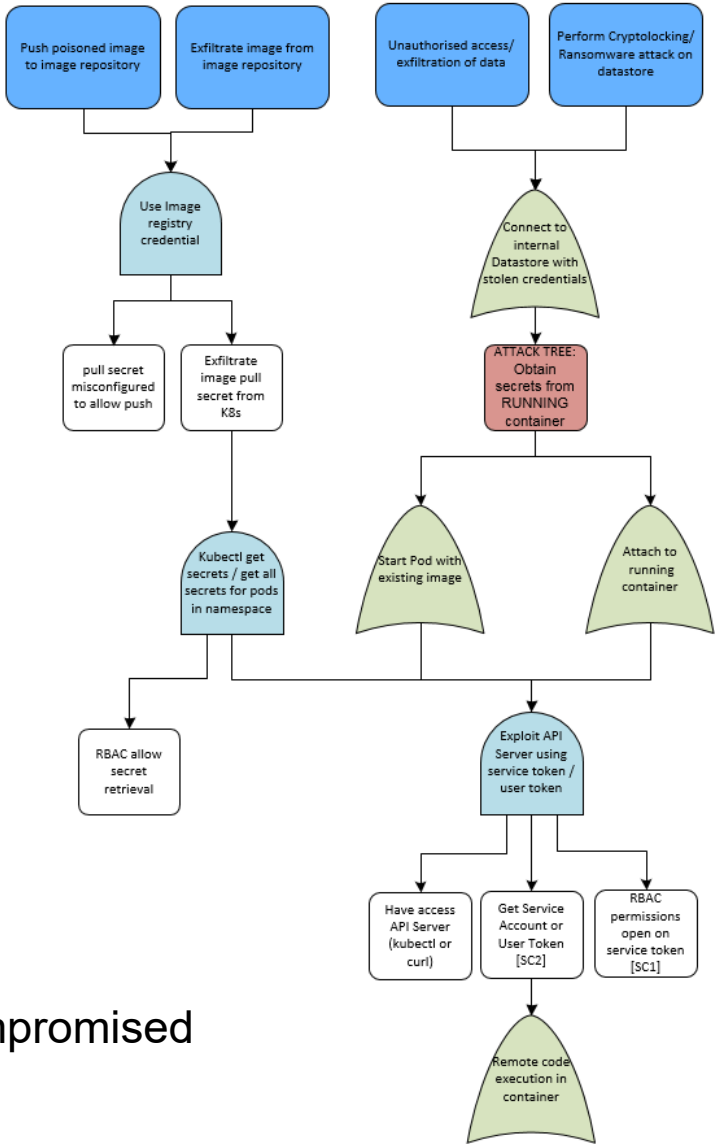
- Establish Persistence – container compromise



Kubernetes Threat Model



Scenarios – compromised container



Kubernetes Security

K8s Security Features

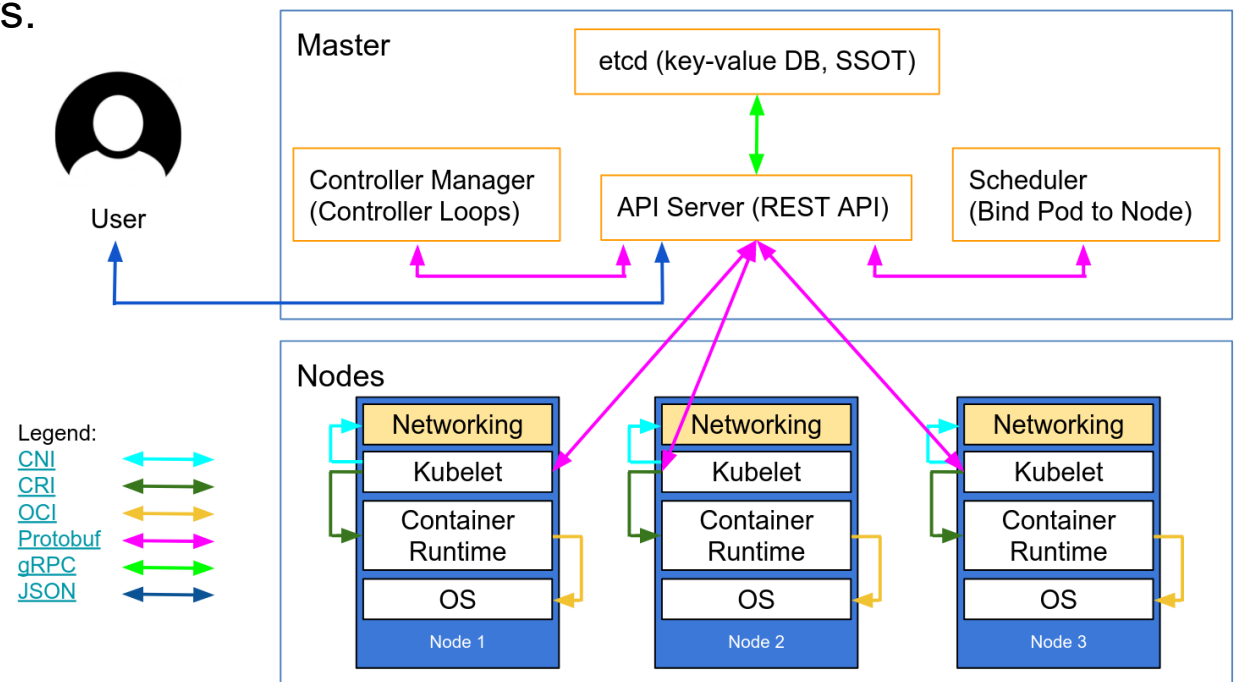
- **Authentication:** Many ways to authenticate, e.g., client certs, OpenID, static token file, etc.
- **Authorization:** RBAC, ABAC, Node, and Webhook
- **Audit Logging:** Periodically watch audit logs
- **Network Policies:** All pods can talk to others by default. Add network policy.
- **Pod Security Policies:** Add pods security policies.
- **Kubernetes Secrets:** Use kubernetes secrets instead of config maps to store sensitive data.

Kubernetes Cluster: Best Practices

Reduce the Attacks Surface for: Host, Container Images, and Cluster

- Minimize privilege to applications running on host.
- Know your base image when building containers.
- The smaller the image, the better
- Don't rely on :latest tag
- Check for vulnerabilities periodically
- Don't run as root
- Limit host mounts

- Ensure TLS checklist for cluster



Kubernetes Security: Open-source Tooling



Checks whether Kubernetes
is securely deployed



Security risk analysis for
Kubernetes resources



Static analysis of
vulnerabilities in containers



Kubescape

Security platform that scans clusters, kubernetes manifest
files, code repos, container registries and images



- CLI tool to sign and verify artifacts
- Solution to secure updates and distr.
- Used in Docker Trusted Registry

#sdukd

SDU 

<https://landscape.cncf.io/>

Reference

- [Kubernetes Security Cheat Sheet](#)
- [Kubernetes Threat Model – CNCF Financial User Group](#)
- [Kubernetes Threat Model – CloudSecDocs](#)
- [A Deep Dive into Kubernetes Threat Modeling – Trend Micro](#)
- [Kubernetes Hardening Guidance \(DoD\)](#)

Revision

- Selected topics
- Clarification / Confusion on any topic?
- Questions...

Introduction
Vulnerability Assessment
Penetration Testing
Social Engineering, Phishing
Threat Modeling I
Threat Modeling II
IDS, Malware, DoS, Crypto, MPC

Mock Test

→ **Password for Mock Test:** CSF26MOCKTEST

→ **Duration:** 45 min

→ **Total Number of Questions:** 15

(10 multiple choice and 5 short essay)

Exam Guidelines

- **The exam takes place on campus-** check your allocated room on the list at DE on the day before the exam.
- **The exam contains MCQ/MAQ and short-answer questions.**
- **You can freely navigate between questions.**
- **You can select multiple options as some questions have more than one options that are correct.**

Exam Question Paper Guidelines

 **Type of Questions:** MCQ/MAQ and Short Essay/Answers Questions

 **Total Number of Questions:** 50

 **MCQ/MAQ:** 35 Questions and **Short Answers:** 15 Questions

 **Total Time Duration:** 2.5 Hours & 3 Hours (with extra time)

Exam Preparation Guidelines



Learn and Understand the Concepts in Slides & Exercises



Focus on Lecture Slides and Exercises



Practice the Exercises for Labs 2, 3, 5, 6, 8, and 9

Any other Questions?



Feel free to reach out via email: saa@mmmi.sdu.dk or him@mmmi.sdu.dk

GOOD LUCK

On Your End Semester Examinations!

Every exam is a step closer to your goal. Embrace the journey with optimism and confidence. Aim high, stay focused and success will be within your reach.

Thank You!